

MESOSCALE MICROSTRUCTURE MODELING (MS5)

PROBLEM STATEMENT

Active Parameter Tracking in Grain Growth

Team Members:

Vishnu Prasannan (MS22BTECH11030)

Ashish Bhardwaj (MS22BTECH11007)

Instructor:

Prof. Saswata Bhattacharya

April 11, 2025

Grain Growth

Governing Equations

Allen-Cahn Equation

$$\frac{\partial \phi_i}{\partial t} = -\mathcal{L}_i \cdot \frac{\delta F}{\delta \phi_i} \quad i = 1, 2, \dots, N \quad (1)$$

Here, F represents the total free energy of the system, consisting of bulk energy density f and the gradient energy penalty associated with the interfaces of different grains.

$$F = \int_V \left[f(\phi_1, \phi_2, \dots, \phi_N) + \sum_i^N \frac{\kappa_i}{2} |\nabla \phi_i|^2 \right] dv \quad (2)$$

The bulk energy density function f is modeled using a multi-well potential for each grain order parameter ϕ_i , penalizing overlaps between grains through the interaction term with coefficient γ

$$f(\phi_1, \phi_2, \dots, \phi_N) = \sum_{i=1}^N \left(-\frac{1}{2} \alpha \phi_i^2 + \frac{1}{4} \beta \phi_i^4 \right) + \gamma \sum_{i=1}^N \sum_{j \neq i}^N \phi_i^2 \phi_j^2 \quad (3)$$

Taking the functional derivative of F with respect to gives us the driving force for the evolution of each grain, combining local bulk terms and gradient penalty terms.

$$\frac{\delta F}{\delta \phi_i} = \frac{\partial f}{\partial \phi_i} - \kappa_i \nabla^2 \phi_i \quad (4)$$

Computing the partial derivative of f with respect to, we obtain contributions from self-interaction and interactions with neighboring grains.

$$\frac{\partial f}{\partial \phi_i} = -\alpha \phi_i + \beta \phi_i^3 + 2\gamma \phi_i \sum_{j \neq i}^N \phi_j^2 \quad (5)$$

Substituting this into the expression for $\frac{\delta F}{\delta \phi_i}$, we get the explicit form of the variational derivative governing the evolution of ϕ_i .

$$\frac{\delta F}{\delta \phi_i} = -\alpha \phi_i + \beta \phi_i^3 + 2\gamma \phi_i \sum_{j \neq i}^N \phi_j^2 - \kappa_i \nabla^2 \phi_i \quad (6)$$

Finally, substituting into the governing equation (Eq 1), we obtain the time evolution equation for each order parameter which is a multivariate Allen-Cahn type equation.

$$\frac{\partial \phi_i}{\partial t} = -\mathcal{L}_i \left(-\alpha \phi_i + \beta \phi_i^3 + 2\gamma \phi_i \sum_{j \neq i}^N \phi_j^2 - \kappa_i \nabla^2 \phi_i \right) \quad i = 1, 2, \dots, N \quad (7)$$

Finite Element Method

$$\frac{\phi_i^{n+1} - \phi_i^n}{\Delta t} = -\mathcal{L}_i \left(-\alpha \phi_i^n + \beta (\phi_i^n)^3 + 2\gamma \phi_i^n \sum_{j \neq i}^N (\phi_j^n)^2 - \kappa_i \nabla^2 \phi_i^n \right) \quad i = 1, 2, \dots, N \quad (8)$$

Fourier Spectral Method

From equation 1:

$$\frac{\partial \phi_i}{\partial t} = -\mathcal{L}_i \frac{\partial f}{\partial \phi_i} + \mathcal{L}_i \kappa_i \nabla^2 \phi_i \quad i = 1, 2, \dots, N \quad (9)$$

Applying Fourier transform to both sides of Eq. 1, the Laplacian term $\nabla^2 \phi_i$ transforms to $-(2\pi k)^2 \tilde{\phi}_i$ in Fourier space.

$$\frac{\partial \tilde{\phi}_i}{\partial t} = -\mathcal{L}_i \frac{\partial \tilde{f}}{\partial \phi_i} + (2\pi i k)^2 \mathcal{L}_i \kappa_i \tilde{\phi}_i \quad (10)$$

Using a forward difference scheme for the time derivative, the discretized form of the equation in Fourier space becomes:

$$\frac{\tilde{\phi}_i^{n+1} - \tilde{\phi}_i^n}{\Delta t} = -\mathcal{L}_i \frac{\partial \tilde{f}^n}{\partial \phi_i} - 4\pi^2 k^2 \mathcal{L}_i \kappa_i \tilde{\phi}_i^{n+1} \quad (11)$$

Defining $\tilde{g}(\phi_i)$ as the Fourier transform of the bulk free energy derivative evaluated at time step n :

$$\tilde{g}(\phi_i) = \frac{\partial \tilde{f}^n}{\partial \phi_i} \quad (12)$$

Rearranging the terms, the explicit expression for $\tilde{\phi}_i^{n+1}$ in Fourier space is obtained as:

$$\tilde{\phi}_i^{n+1} = \frac{\tilde{\phi}_i^n - \Delta t \mathcal{L}_i \tilde{g}(\phi_i)^n}{1 + 4\pi^2 \mathcal{L}_i \kappa_i k^2 \Delta t} \quad (13)$$

1. Evaluate $\frac{\delta f}{\delta \phi_i}$ using the current time step value of ϕ_i .
2. Fourier transform $\frac{\delta f}{\delta \phi_i}$ and ϕ_i .
3. Calculate the spatial variation of ϕ_i in Fourier space for the time $t + \Delta t$, using Eq. 12.
4. Using inverse Fourier transformation, bring back the results to real space for the solution at $t + \Delta t$.

Active parameter tracking (APT)

APT is a clever trick to reduce computation by only focusing on the active grains at each grid point. At each grid point $\mathbf{x}$ and time t , defining the **active set**:

$$P(\mathbf{x}, t) = \{i, \phi_i : |\phi_i(\mathbf{x}, t)| > \epsilon\} \quad (14)$$

where ϵ is a small positive threshold.

Since grains migrate over time, some grains may enter or exit a grid point. the union of P from neighboring points (1st and 2nd neighbors) and call it $Q(\mathbf{x}, t)$:

$$Q(\mathbf{x}, t) = \bigcup_{p,q=0}^1 P(\mathbf{x} + p\mathbf{e}_1 + q\mathbf{e}_2, t) \quad (15)$$

The number of active parameters is:

$$Q_a(\mathbf{x}, t) = |Q(\mathbf{x}, t)| \quad (16)$$

where $|s|$ means the cardinality of the set s .

Allen-Cahn equation from equation 1, for each order parameter ϕ_i is given by:

$$\frac{\partial \phi_i}{\partial t} = -L_i \frac{\delta F}{\delta \phi_i}$$

Substituting the functional F from equation 6, this can expand to:

$$\frac{\partial \phi_i}{\partial t} = -L_i \left(-\alpha \phi_i + \beta \phi_i^3 + 2\gamma \phi_i \sum_{\substack{j=1 \\ \xi_j \neq i}}^{Q_a} \phi_{\xi_j}^2 - \kappa_i \nabla^2 \phi_i \right) \quad (17)$$

$$\forall (i, \phi_i) \in Q(\mathbf{x}, t) \quad \text{and} \quad (\xi_j, \phi_{\xi_j}) \in Q(\mathbf{x}, t), \quad j = 1, \dots, Q_a.$$